Solution

Let the vertex be $(0,0)$.

The equation of the parabola is

$y^2 = 4ax$

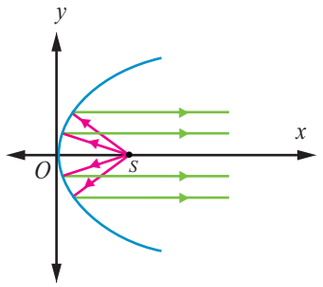


Fig. 5.60

(1) Since the diameter is $40\,cm$ and the depth is $30\,cm$, the point $(30,20)$ lies on the parabola.

$$20^2 = 4a \times 30$$

$$4a = \frac{400}{30} = \frac{40}{3}.$$

Equation is $y^2 = \frac{40}{3}x$.

(2) The bulb is at focus $(a,0)$. Hence the bulb is at a distance of $\frac{10}{3}cm$ from the vertex.

Example 5.37

An equation of the elliptical part of an optical lens system is $\frac{x^2}{16}+\frac{y^2}{9}=1$. The parabolic part of the system has a focus in common with the right focus of the ellipse .The vertex of the parabola is at the origin and the parabola opens to the right. Determine the equation of the parabola.

Solution

In the given ellipse $a^2 = 16,\ b^2 = 9$

then $c^2 = a^2 - b^2$

$c^2 = 16 - 9$

$= 7$

$c = \pm\sqrt{7}$

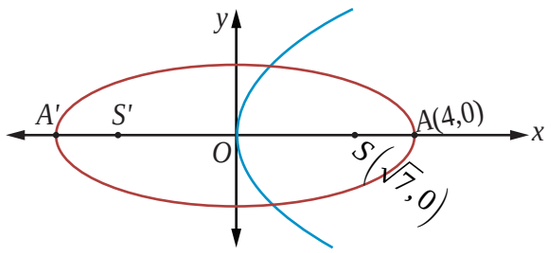


Fig. 5.61

Therefore the foci are $F\left(\sqrt{7},0\right), F'\left(-\sqrt{7},0\right)$. The focus of the parabola is $\left(\sqrt{7},0\right) \Rightarrow a = \sqrt{7}$.

Equation of the parabola is $y^2 = 4\sqrt{7}x$.

5.7.5 Reflective Property of an Ellipse

The lines from the foci to a point on an ellipse make equal angles with the tangent line at that point (Fig. 5.62).

The light or sound or radio waves emitted from one focus hits any point P on the ellipse is received at the other focus (Fig. 5.63).

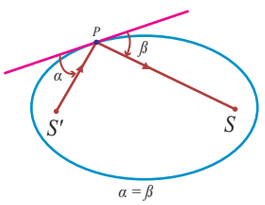


Fig. 5.62

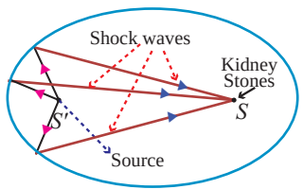


Fig. 5.63

Example 5.38

A room $34m$ long is constructed to be a whispering gallery. The room has an elliptical ceiling, as shown in Fig. 5.64. If the maximum height of the ceiling is $8m$, determine where the foci are located.

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